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Organic Agriculture Production of Traditional Rice Varieties – Code of Practice

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Foreword

In 2021, the Department of Agriculture Regional Field Office - Cordillera Administrative Region (DA RFO - CAR) requested to develop a PNS on the COP for Organic Agriculture Production of Traditional Rice Varieties to improve the production quality of traditional rice varieties and ensure their organic integrity. A Technical Working Group (TWG) was created through Special Order No. 817, series of 2021 (Addendum to Special Order 81, series of 2021 entitled, “Creation of TWG for the development of PNS for agriculture and fishery products, machinery and equipment”). The TWG was composed of representatives from the relevant government agencies, academe, research institutions, and civil society organizations. The draft standard was presented to the relevant stakeholders during the initial stakeholder consultations held March 10 to 11 and 18 to 19, 2021 and the final stakeholder consultation held August 24, 2021. It was finalized by the TWG before it was endorsed to the DA Secretary for approval.

This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2.

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1 Scope

This standard establishes the recommended practices for the organic pre-production, production, and post-production of traditional rice (*Oryza sativa* L.) varieties intended for human consumption. It provides the steps in the production of traditional rice varieties that will ensure the organic integrity and protect the cultural heritage of the product.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the cited edition applies. For undated references, the latest edition of the referenced document (including any amendments) applies:

Bureau of Agriculture and Fisheries Standards (BAFS) – Department of Agriculture (DA). (2016). Philippine National Standards (PNS) for organic agriculture (PNS/BAFS 07:2016). http://www.bafs.da.gov.ph/bafs_admin/admin_page/pns_file/2021-03-01-BAFS%20PNS%2007-2016%20Organic%20Agriculture.pdf

BAFS – DA. (2020). National list of permitted substances for organic agriculture (Department Circular No. 09, series of 2020). http://www.bafs.da.gov.ph/permitted_substances_oad

BAFS – DA. (2021). Registered Organic Soil Amendment (OSA) producers. http://www.bafs.da.gov.ph/organic_osa

BAFS – DA. (2021). Registered Organic Bio-Control Agent (OBCA) producers. http://www.bafs.da.gov.ph/organic_obca

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply:

3.1
buffer zone
clearly defined and/or identifiable boundary area bordering an organic production site that is established to limit application of, or contact with, prohibited substances from an adjacent area

3.2
community registration
registration of the relevant information regarding the seeds and planting materials to the competent authority

3.3
contamination
contact of organic crops, land or products to substance that would compromise their organic integrity

3.4
conventional
any material, production, or processing practices that is not certified organic or “in-conversion”

3.5
conversion period (transition period)
time between the start of organic management and certification of the crop or production system or site as organic

3.6
degree of milling
extent to which the bran layers have been removed in hulled rice

3.6.1
fully-polished rice
overmilled rice (OMR)
rice kernel from which the entire hull, germ, bran layers, and some parts of the endosperm have been removed

3.6.2
semi-polished rice
rice kernel from which the hull, the germ, the outer bran layer, and the greater part of the inner bran layers have been removed but parts of the bran layer lengthwise steaks in the dorsal portion of the kernel remain

3.6.2.1
undermilled rice (UMR)
rice kernel from which the hull, the germ, the outer bran layer, and the greater part of the inner bran layers have been removed but parts of the lengthwise streaks of the bran layers remain (in the dorsal portion) in more than 40% of the kernels

3.6.2.2
regular milled rice (RMR)
rice kernel from which the hull, the germ, the outer bran layers, and the greater part of the inner bran layers have been removed but parts of the lengthwise streaks of the bran layers remain (in the dorsal portion) in 20% to 40% of the kernels

3.6.2.3
well milled rice (WMR)
rice kernel from which the hull, the germ, the outer bran layers, and the greater part of the inner bran layers have been removed, but parts of the lengthwise streaks of the bran layers remain (in the dorsal portion) in 1% to 19% of the kernels

3.7
dibbling
practice of sowing seeds in a hole bored (dibble) in the soil

3.8
drying
process of removing excess available water from grain through evaporation by the application of heat

3.9
farm structure
common agricultural structures associated with agricultural use such as harvesting, growing, and storage of crops

3.10
harrowing
practice of breaking up and smoothing out the surface of the soil and is usually done after plowing

3.11
harvesting
process of collecting the mature rice crop from the field. *Palay* harvesting activities include reaping, stacking, handling, threshing, cleaning, and hauling. These can be done individually, or a combine harvester can be used to perform the operations simultaneously.

3.12
heirloom rice
type of traditional rice variety that are handed down by several generations through family members that are normally grown by small landholders. It is considered as the highest-quality traditional rice varieties grown in provinces such as in Abra, Kalinga, Benguet, Mt. Province, Ifugao, Apayao, North and South Cotabato, Oriental Mindoro, Negros Occidental, Negros Oriental, Sultan Kudarat, Bukidnon, and Misamis Oriental.

3.13
hauling
activity of transport (by foot or vehicle) of the harvest

3.14
hook
curved hand tool that is used for securing and moving loads

3.15
leveling
practice of smoothing the surface of the soil ensuring that the land is of the same height

3.16
organic agriculture
holistic production management system which promotes and enhances agro-ecosystem health, including biodiversity, biological cycles, and soil biological activity; emphasizes the use of management practices over the use of off-farm inputs; and utilizes cultural, biological, and mechanical methods as opposed to synthetic materials. Organic agriculture combines tradition, innovation, and science to benefit the shared environment and promote fair relationships and a good quality of life for all involved.

3.17
organic integrity
adherence to the principles, objectives, and standards for organic production

3.18
organic soil amendments
include all the products (e.g. organic fertilizer, organic soil conditioner, microbial inoculant, and organic plant supplement) covered by the scope of PNS BAFS 183:2020 (PNS of organic soil amendments)

3.19
postharvest
process of collecting or separating grain from its site of immediate growth or production and subject to further processing of post-production stages of drying, milling, storage, and marketing

3.20
post-production
series of activities that grain crops undergo which include harvesting, threshing, hauling, drying, milling, handling, packaging, and storage

3.21
pre-production
series of activities before planting or sowing of grain crops which include land preparation, preparation of planting materials, soil nutrient management and soil conditioning, and field cleaning and sanitation

3.22
production
series of activities from planting to before harvest which include water management, integrated pest management, nutrient management, and field cleaning and sanitation

3.23
production site
land or area under the control of the farmer or group of farmers, including all farming activities or enterprises

3.24
purging
process of completely cleaning or removing remaining *palay* and/or unpolished/semi-polished rice stuck inside the combine harvester, drying, or milling machine before the start of a specific post-production operation

3.25
sanitation
practice which involves the removal and/or destruction of weeds and potential sources of diseases and pest infestation from the field

3.26
synchronous planting
practice of planting within one-month period covering adjacent rice fields within an area

3.27
traditional rice varieties (landraces)
rice cultivars that are indigenous to an area or landraces considered as a rich source of genetic diversity due to its health and nutritional values, grain quality, texture, aroma, and resistance to biotic and abiotic stresses

3.28
trampling
practice of treading on and crushing the soil manually or with an assistance of animals in order to incorporate plant residues and control weeds

3.29
threshing
process of removing paddy from the panicles

3.30
unpolished rice
brown rice
cargo rice
dehulled rice
husked rice
pinawa
palay from which only the hull has been removed and with the bran layer still intact

4 Minimum requirements

4.1 Conversion period (transition period)

The minimum requirements for the conversion period to organic agriculture production of traditional rice varieties should be in accordance with the PNS on Organic Agriculture (PNS/BAFS 07:2016).

4.2 Farm location

4.2.1 The production site and adjoining areas surrounding the site or farms should be evaluated for their suitability for agricultural land use. It is necessary to obtain a history of prior land use (e.g. sanitary landfill, cemetery, industrial plants, mining, etc.) in order to identify potential hazards specifically chemical (e.g. heavy metals) and physical hazards (e.g. broken glass, plastics, etc.).

4.2.2 The evaluation of the suitability of the farm location should also include an assessment of the adjoining crops and type and degree of fertilizer usage.

4.2.3 Production sites situated near urban and industrial areas should be thoroughly assessed by the competent authority for possible contamination.

4.2.4 The farmer should implement appropriate preventive or mitigating measures against potential hazards such as those identified in Annex A (Some recommended phytoremediation for heavy metal contamination in soil and water).

4.3 Farm environment

4.3.1 The production, post-harvest, and storage areas should be kept clean and organized at all times. Field cleaning should always be practiced.

4.3.2 Proper waste disposal of biodegradable and non-biodegradable materials should be observed.

4.4 Soil and soil nutrients

4.4.1 Prior to land preparation, the soil samples should be gathered and analyzed for appropriate organic soil amendment recommendation towards balanced nutrient application. Soil analysis should be done at least every three years by the competent authority or their recognized laboratories.

4.4.2 For areas prone to heavy metal contamination, the soil should be analyzed annually by the competent authority or their recognized laboratories. Maximum Levels (MLs) for heavy metals should not exceed the safety limits set by the competent authority.

4.4.3 Some recommendations on the preventive or mitigating measures for heavy metal contamination in soil are indicated in Annex A (Some recommended phytoremediation for heavy metal contamination in soil and water).

4.5 Irrigation water

4.5.1 Water source(s) for farm operations should be identified for possible contamination.

4.5.2 Water for irrigation should be analyzed for water quality from annually to every three years depending on the assessment of the competent authority on the perceived hazards or contaminants. The testing should be done by the competent authority or their recognized laboratories.

4.5.3 Some recommendations on the preventive or mitigating measures for heavy metal contamination in water are indicated in Annex A (Some recommended phytoremediation for heavy metal contamination in soil and water).

4.6 Farm structure and maintenance

4.6.1 Farms structures should be appropriately designed for the intended purpose and constructed separately from one another to minimize contamination.

4.6.2 There should be a separate or centralized structure for the following:

a) storage areas for organic fertilizers, biocontrol agents, other farm supplies, materials, and tools;
b) shed for the machineries available;
c) storage facility for safekeeping of the produce; and
d) toilet facilities for farm workers that are located in areas where it will not contaminate production site and the crops.

4.6.3 The farmer should ensure that the harvested rice should only have contact with permitted substances indicated in Department Circular No. 09, series of 2020 (National list for permitted substances for organic agriculture) if the farm utilizes their home for storage.

4.6.4 All farm structures, facilities and equipment should always be kept clean and well-maintained for optimal operating conditions.

4.6.5 Sewerage, waste disposal, and drainage systems should be appropriately constructed to minimize the risk of contaminating the production areas and water supplies with hazardous chemicals such as pesticides, heavy metals, and biological hazards (e.g. pathogens from raw manure or crop residues).

4.6.6 Irrigation water ways should be maintained to provide effective delivery of water.

4.6.7 Stray animals should not be allowed into or kept in production areas with standing crops.

4.6.8 Animal proofing and implementation of adequate pest control measures should also be implemented in storage and packing areas.

4.6.9 Toilet facilities should be provided for stay-in farm workers and should be properly cleaned and maintained. These should not be located close to a water source or in places where these could cause contamination.

4.7 Farm practices

4.7.1 Land preparation

4.7.1.1 Proper land preparation based on the contour, soil type, and rainfall pattern in the production area in various rice ecosystems should be observed. This will ensure healthy and uniform plant growth, conserve or improve soil physical condition, and provide effective weed control measures.

4.7.1.2 For terraces:

a) field cleaning should be done throughout the cropping period;
b) for flat areas, plowing and repeated harrowing may be done to ensure good tilth, uniform soil and fine in texture, which promote good plant growth and effective weed control;
c) flooding should be done if the soil is too dry;
d) contour plowing and terracing are recommended in areas where tillage is practiced;
e) dikes, ripraps, and canals should be repaired for efficient irrigation;
f) trampling should be conducted with plant residues and organic fertilizers trampled 10 cm to 15 cm deep. It is recommended to incorporate green manure (such as legumes, *Azolla,* wild sunflower) in trampling; and
g) leveling should be practiced to ensure uniform water distribution and pest and weed management. There should be shallow water level at 2 cm to 5 cm and there should be no visible mound above the water surface for effective weed control.

4.7.1.3 For flat (0 to 3% slope), slightly rolling (>3% and <18% slope), to steep hills and 18% above slope:

a) the farmer should conduct a cultural weed control practice (e.g. plowing and trampling) depending on the weed population level;
b) in areas where tillage is not practiced, the farmer should employ dibbling or transplanting method of seeding; and
c) harrowing may be done once or depending upon the status of the soil (e.g. well pulverized, weed-free, ready for planting).

4.8 Planting and seed materials

4.8.1 For heirloom rice, the farmer should use community registered seed varieties approved by the competent authority and adapted in the locality.

4.8.2 For traditional rice varieties, the farmer should use pure seeds approved by the competent authority and adapted in the locality.

4.8.3 The farmer should record the sources of seed materials including product identity (e.g. company name, lot number, variety, germination percentage, date tested, yield potential, and maturity).

4.8.4 The farmer should select and harvest panicles that are disease free, mature, and of the same variety for seeds.

4.8.5 The seeds for storage should be dried gradually to less than 12% moisture content and stored in an airtight condition free from pests.

4.8.6 The farmer should follow the recommended seeding rate per hectare, size of seedbed area, number of seedlings per hill, age of seedlings, and appropriate planting distance.

4.8.7 Seed soaking and incubation should be practiced for 24-36 hours each before seed sowing.

4.9 Integrated Nutrient Management

4.9.1 The recommended application rate of organic soil amendments, such as organic fertilizer, should be based on any or combination of the following methods of determining soil nutrient deficiency: laboratory analysis, Minus-One Element Technique (MOET) (before transplanting), and Leaf Color Chart (LCC).

4.9.2 As part of land preparation, the farmer should:
a) incorporate rice straws, other organic materials, or crop residues;
b) avoid burning of rice straws; and

c) use compost fungus (such as products containing *Trichoderma harzianium*) and beneficial bacteria (such as Effective Microorganism 1 or EM1 products) to hasten the decomposition of rice straw.

4.9.3 The farmer should only use substantially decomposed organic materials except for those used as green manure. Raw and/or partially decomposed manures and other farm wastes should be confined in a designated area to allow substantial decomposition prior to utilization. On-farm produced organic soil amendments shall adhere to the list of permitted substances and materials in the Department Circular No. 09, Series of 2021 (National list of permitted substances for organic agriculture).

4.9.4 The farmer should observe appropriate method and time of application of the recommended combination and amount of organic soil amendments based on the results of soil and plant-based analysis. Commercially-produced organic soil amendment registered by the competent authority should be used accordingly.

4.9.5 The farmer should store organic soil amendments separately in a clean and dry area (preferably slightly elevated above ground on pallets).

4.9.6 The farmer should isolate storage area of organic soil amendments from organic traditional/heirloom rice drying and storage areas to prevent contamination due to leaching, runoff, or wind drift.

4.9.7 The farmer should keep a complete set of records of organic soil amendments and organic soil amendments preparations. Information includes source of organic materials, details of the composting procedures, dates, amounts, and methods of applying the organic soil amendments as well as the person responsible for the application.

4.10 Integrated pest management

4.10.1 A pest management program should be followed to manage rice pests. The farmer should:

a) ensure good land preparation and crop establishment;
b) use varieties that are resistant (if any) to major insect pests and diseases prevalent in the locality;
c) use good quality seeds free from pests and contaminants, plant healthy seedlings, and follow good agronomic practices;
d) follow synchronous planting scheme after a fallow period;
e) perform cultural practices that are favorable to the conservation of beneficial insects;
f) practice varietal and/or crop rotation to minimize build-up of insect pests and diseases;
g) conduct regular monitoring of crops to assess the prevalence of pests and for timely farm management interventions; and

h) Use registered biocontrol agents at different critical stages as an option. Only biocontrol agents indicated in the list of registered OBCA producers are allowed. Proper observance of pre-harvest intervals must be maintained.

4.11 Weed management

The farmer should practice appropriate weed control measures.

4.12 Water management

4.12.1 The farmer should maintain the recommended water requirement at different growth stages to avoid moisture stress and excessive water particularly during flowering up to the maturation stage.

4.12.2 Field drainage may be done regularly. Drainage canals of at least 25 cm wide and 5 cm deep should be constructed for proper drainage.

4.12.3 Unless alternate wetting-drying water management is practiced, the following should be followed:

4.12.3.1 For direct seeded:

a) 0 cm to 1 cm water depth is recommended from sowing until 30 Days After Seeding (DAS);
b) 2 cm to 3 cm water depth is recommended from 30 DAS to flowering; and
c) 3 cm to 5 cm water depth is recommended from flowering until two weeks before harvesting.

4.12.3.2 For transplanted:

a) 0 cm to 1 cm water depth is recommended from transplanting until 5-10 Days After Transplanting (DAT);
b) 2 cm to 3 cm water depth is recommended from 5-10 DAT until flowering; and
c) 3 cm to 5 cm water depth is recommended from flowering until 1-2 weeks before harvesting.

4.12.4 The farmer should follow other recommended cultural practices of rice, including maintenance of the recommended row and plant spacing to avoid overcrowding.

4.12.4.1 For transplanted rice:

a) For highland production, it is recommended that 35-46 days old seedlings will be used for transplanting;
b) For lowland production, it is recommended that 18-21 days old seedlings will be used for transplanting;
c) There should be 2-3 seedlings per hill at a minimum planting distance of 20 cm by 20 cm spacing or whatever spacing is recommended in the locality; and
d) Missing hills should be replanted within seven days after transplanting.

4.12.4.2 For wet bed method:

a) For a 1,000 m^2 production area, seeds should be planted with 3-5 inches interval between panicles;
b) For broadcasting, 6 kg of pre-germinated seeds should be broadcasted per 60 m^2 seedbed area;
c) Pre-germinated seeds should be used; and
d) The soil should be saturated within 5-7 days after sowing. The water level should be increased to 2 cm until seed puling.

4.12.4.3 For direct dry-seeded rice:

a) For a 1,000 m^2 production area, 6 kg of seeds should be broadcasted evenly in 60 m^2 seedbed area;
b) The seeds should be covered with fine soil, rice straw or carbonized rice hull; and
c) The soil should be kept saturated until seed pulling.

4.12.5 The farmer should conduct regular monitoring at all crop stages and provide appropriate measures to address problems that may arise.

4.13 Harvesting

4.13.1 There should be no sources of contamination that should affect the quality of organic traditional rice/heirloom *palay*.

4.13.2 Harvesting of organic traditional/heirloom *palay* should not coincide with the harvesting of conventional *palay* adjacent to the farm. *Palay* harvested in the designated buffer zones should be considered non-organic.

4.13.3 Harvesting should be done when 80% to 85% of the panicle are mature for shattering varieties. For non-shattering varieties, ripening can be extended up to 90% to 100%.

4.13.4 For stored organic traditional/heirloom *palay*, harvested panicles should be threshed immediately 1 day (during wet season) or 2 days (during dry season) after harvesting.

4.13.5 Combine harvesters, reapers, threshers, scythes, and other tools and accessories should be thoroughly cleaned before threshing. If the equipment is not exclusively used for organic traditional/heirloom *palay*, purging or flushing should be practiced. The recommended amount of the purged or flushed *palay* to be segregated depends on the capacity of the combine harvesters, reapers, threshers, and scythes. Segregated *palay* from the thresher should be considered as conventional.

4.13.6 Brand new and clean recycled sacks should be used recommended for traditional/heirloom *palay*.

4.13.7 Sacks should be properly labeled to avoid mixing of different varieties of *palay* and organic and conventional *palay*.

4.14 Hauling (farm to dryer)

4.14.1 Organic traditional/heirloom *palay* and conventionally grown *palay* should be hauled separately. Hauling for different varieties should also be done separately.

4.14.2 The farmer should haul newly harvested organic traditional/heirloom *palay* immediately after harvest.

4.14.3 Hauling facilities to be used for collecting and transporting the harvested organic traditional/heirloom *palay* from the farm should be clean and dry.

4.15 Drying

4.15.1 Organic traditional/heirloom and conventionally grown *palay* should be dried separately.

4.15.2 Mechanical drying is recommended in drying organic traditional/heirloom *palay*. Clean mechanical dryers should be used. If possible, a mechanical dryer should be dedicated solely for drying organic traditional/heirloom *palay*. If not possible, organic traditional/heirloom *palay* should be dried first before conventionally grown *palay*. However, if the mechanical dryer was previously used for conventional rice and/or different organic traditional/heirloom varieties, purging or flushing should be practiced. The amount of the purged or flushed *palay* to be segregated depends on the capacity of the dryer. Segregated rice from purging or flushing is considered as conventional.

4.15.3 Sun drying is allowed as long as the use of drying equipment such as solar bubble dryer is applied. However, drying in highways and roads is strictly prohibited. Drying facilities like concrete pavement with underlay (*sakolina*, plastic and others) should be properly cleaned and free from oil spill, manure, residues, and other contaminants before drying.

4.15.4 During drying, birds and other stray animals should be kept away from the drying area to avoid contamination of organic traditional/heirloom *palay*. Personnel in charge of drying organic traditional/heirloom *palay* should observe proper hygiene within the premises.

4.15.5 Only brand-new and clean recycled sacks should be used.

4.15.6 Proper tagging of sacks should be observed to avoid mixing of organic traditional/heirloom from conventionally grown *palay*.

4.15.7 Dried traditional/heirloom *palay* should have a maximum moisture content of 14% when tested by properly calibrated moisture meters.

4.16 Hauling (dryer to storage)

4.16.1 Organic traditional/heirloom and conventionally grown *palay* should be hauled separately.

4.16.2 Transport facilities should be properly cleaned every time organic traditional/heirloom *palay* are to be hauled.

4.16.3 Hauling facilities to be used for collecting and transporting the dried organic traditional/heirloom *palay* from the dryer to storage should be clean and dry.

4.17 Milling

4.17.1 As far as practicable, a rice mill should be dedicated solely for milling organic traditional/heirloom *palay*.

4.17.2 To avoid contamination, all milling equipment should be properly and thoroughly cleaned before usage.

4.17.3 Organic traditional/heirloom and conventional *palay* should be milled separately. Likewise, different traditional/heirloom rice varieties should be milled separately.

4.17.4 If the rice mill is not exclusively used for organic traditional/heirloom *palay*, organic traditional/heirloom *palay* should be milled first before the conventionally grown *palay*. However, if the mill was previously used for conventional rice and/or different organic traditional/heirloom varieties, purging or flushing should be practiced. The amount of the purged or flushed unpolished/semi-polished rice to be segregated depends on the capacity of the mill. Segregated rice from purging or flushing is considered as conventional.

4.17.5 Only brand-new sacks and other packaging materials should be used. Labeling of sacks and other packaging materials for organic traditional/heirloom unpolished/semi-polished/polished rice and should be in accordance with the requirements of the PNS on Organic Agriculture (PNS/BAFS 07:2016).

4.17.5.1 Dried *palay* shall be packed in new food-grade packaging materials in accordance with the PNS/BAFS 141:2019 (Code of Good Agricultural Practice [GAP] for rice).

4.17.5.2 Suitable food-grade ink shall be used for printing or labeling the packaging material.

4.17.5.3 The labeling should comply with the relevant regulations on organic agriculture by the competent authority.

4.17.5.4 The minimum requirements for labeling of unpolished/semi-polished rice shall be:

a) Name of the product (e.g. *palay*) if the contents are not visible from the outside;
b) Variety of the *palay;*
c) Degree of milling (unpolished and semi-polished – undermilled, regular milled, well milled);
d) Moisture content, in percent;
e) Net weight (shall be declared in 'nominal net weight' in SI unit, e.g. kg); and
f) Name and address of the distributor.

4.18 Hauling and storage facility of organic traditional/heirloom unpolished/ semi-polished rice

4.18.1 The storage areas and hauling facilities should be thoroughly cleaned and free from pests. Cleaning materials should be in accordance with the latest version of Department Circular No. 09, series of 2020 (National list for permitted substances for organic agriculture).

4.18.2 Use of permitted organic based products (e.g. botanical sprays) should be in accordance with the latest version of Department Circular No. 09, series of 2020 (National list for permitted substances for organic agriculture). These products, though optional, are allowed but should not be sprayed directly to the organic traditional/heirloom unpolished/semi-polished rice. Operators should always make a record on the amount and kind of permitted organic-based products used.

4.18.3 Traditional/heirloom organic unpolished/semi-polished rice should be stored separately from conventional milled rice. If not possible, physical separation should be done.

4.18.4 There should be a separate storage for permitted organic-based products and non-permitted and hazardous products.

4.18.5 In case there is suspicion of traces of pesticides in the storage facilities, these should be removed by using the appropriate cleaning agent and water prior to storage.

4.19 Hauling of organic traditional/heirloom unpolished/semi-polished rice (from mill to storage)

4.19.1 Vehicles and other equipment used during hauling should be properly cleaned prior to loading of organic traditional/heirloom unpolished/semi-polished rice. Use of clean pallets during transport should be observed. Chemically-treated pallets should not be used.

4.19.2 Organic traditional/heirloom unpolished/semi-polished rice should not be hauled together with potential sources of physical, chemical, and biological contaminants.

4.19.3 During hauling, clean tarpaulins and covering materials should be used.

4.20 Storage

4.20.1 The use of "hooks" is prohibited.

4.20.2 Organic traditional/heirloom unpolished/semi-polished/polished rice should be stored at a maximum of 14% moisture content (MC).

4.20.3 Organic traditional/heirloom, unpolished/semi-polished rice should be properly piled and labeled separately from conventional milled rice.

4.20.4 First-In First-Out (FIFO) principle should be observed.

4.20.5 Chemically treated pallets should not be used in the warehouse.

4.21 Pest management in storage

4.21.1 Cultural practice such as daily sanitation as a first line of defense and physical control measures should be encouraged to be implemented as part of the organic integrated pest management approaches.

4.21.2 Use of organic-based products should be in accordance with the latest version of Department Circular No. 09, series of 2020 (National list for permitted substances for organic agriculture).

4.21.3 Pest control measures using chemical pesticides for stored organic traditional/heirloom unpolished/semi-polished rice shall not be allowed.

4.21.4 There should never be direct or indirect contact between organic traditional/heirloom unpolished/semi-polished rice and prohibited substances (e.g. pesticides). When any doubt arises, it should be ensured that no pesticide residues are detected on the organic traditional/heirloom unpolished/semi-milled/polished rice, based on a certificate of analysis issued by the competent authority or laboratories recognized by the competent authority.

4.21.5 Irradiation of the organic traditional/heirloom unpolished/semi-polished rice for pest control shall be prohibited.

4.22 Workers' health, safety, and welfare

4.22.1 Farm workers who are involved in production and post-production activities should wear Personal Protective Equipment (PPE).

4.22.2 Farm workers should be trained and follow the recommended personal hygienic and sanitary practices.

4.22.3 Employers/farm owners should follow the regulatory requirements set by the competent authority for the payment of wages.

4.23 Traceability

4.23.1 Each package/bulk-packed produce leaving the farm should be traceable to the farm/sources.

4.23.2 Records of lot or batch numbers should be maintained for all produce leaving the farm.

4.24 Staff records and training

4.24.1 The farmer should maintain complete personnel and health records of all farm workers.

4.24.2 Staff training records, if any, should be maintained and be readily available.

Annex A
(Informative)

Some recommended phytoremediation for soil and water contamination

The following are recommended plants to be used for phytoremediation (use of plants to extract pollutants from soil and water) for polluted and contaminated soil and water:

A.1 Alpine pennycress (*Thlaspi caerulescens)*

Alpine pennycress (see Figure A.1) member of the Brassicaceae family (cabbage and broccoli family) is a hyperaccumulator, can extract elements from the soil and concentrate it in their plant parts, of zinc (Zn) and cadmium (Cd) (Pirzadah et al., 2015). This plant has genes that allows the roots to regulate the heavy metals taken up in their roots and collect them in their shoots and leaves (Ecosystems Research and Development Bureau [ERDB] – Department of Environment and Natural Resources [DENR], 2010).

Figure A.1. Alpine pennycress plant

A.2 Creamy candles/ creamy stackhousia (*Stackhousia tyronii*)

A known hyper-accumulator of heavy metals, nickel (Ni), creamy stackhousia (see Figure A.2) is an efficient plant to extract heavy metals in contaminated soil from fertilizer and industrial pollution (ERDB -DENR, 2010)

Figure A.2. *Stackhousia* plant

A.3 Sunflower (*Heliantus anuus*)

Figure A.3. Sunflower plant

Sunflower (see Figure A.3), aside from being an ornamental plant, is also known for its emerging potential for phytoremediation. Studies have proven that it is a hyperaccumulator of lead (Pb) and Cd in contaminated soil (Alaboudi et al., 2018). There are also studies that suggest that is also effective for extraction of mercury (Hg) in the soil (Gamage et al., 2021).

A.4 Braken fern (*Pteris vitata*)

Figure A.4. Braken fern plant

Braken fern (see Figure A.4) soaks up arsenic (As) with tremendous efficiency (200 times higher concentration in fern compared to the soil or medium). Based on laboratory tests, the plant consists 2.3% of the plant composition.
Braken fern is known as an efficient hyperaccumulator of arsenic (As). Studies show that there is a 200 times concentration of As in fern compared to the concentration in the soil (ERDB–DENR, 2010).

A.5 Various bamboo species

Figure A.5. *Kawayang tinik* (top left), *kawayang killing* (top right), *anos* (lower left), and *buho* (lower right)

Various bamboo species (see Figure A.5) – *Dendrocalamus asper* (giant bamboo), *bambusa blumeana* (*kawayang tinik*), *Schizostachyum lumampao* (*buho*), *Bambusa vulgaris* (*kawayang killing*), and *Schizostachyum lima* (*anos*)– are being promoted as one of the cost-effective and green hyperaccumulators for heavy metals in soil and water. These can survive and tolerate heavy metal contaminated soil from Pb, Cd, chromium (Cr), and Hg (ERDB-DENR, 2010).

A.6 Kenaf (*Hibiscus cannabinus L.*) and rapeseed (*Brassica napus L.*)

Kenaf, also known as *alas dose*, and rapeseed (see Figure A.6) are known to be effective in phytoremediation of soils and water especially in absorbing selenium (Se). Kenaf has the ability to sanitize oil spills and can also provide additional anchorage to prevent soil erosion (ERDB-DENR, 2010).

Figure A.6. Kenaf plant (left) and rapeseed plant (right)
Retrieved from: https://www.researchgate.net/figure/Hibiscus-Cannabinus-Kenaf-Plant_fig4_309688872

A.7 Vetiver (*Vetiveria zinazoides*)

Figure A.7. Vetiver grass

Vetiver has a great potential for phytoremediation due to its capacity to tolerate toxic levels of heavy metals in soil (Figure A.7). According to studies, its roots could collect more than five (5) times the Cr and zinc (Zn) levels. Studies also suggests that it can gradually reduce high levels of arsenic As, Cd, and Hg. It can also be safe for grazing and mulch due to its exclusion mechanism. Minimal amounts of the heavy metals are translocated in the shoots. (ERDB– DENR, 2010)

A.8 Cogon (*Imperata cylindrica*)

Cogon (see Figure A.8) is proven as excluder of heavy metals Pb, Zn and copper (Cu), with the roots of the species accumulating low levels of metals by avoiding or restricting uptake. Shoots of the species accumulated much lower concentrations of metals by restricting transport because they are symbiotically associated with mycorrhiza.

Figure A.8. Cogon grass on the field

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